EXECUTIVE PITCH & PRESENTATION GUIDE

Retail Customer Intelligence Platform

Comprehensive Project Walkthrough, Strategic Business Case, and Live Demonstration Playbook

Platform: DashFlow / INSIGHT.arc v2.4 **Architecture:** Big Data (Hadoop/Hive/Spark) + ML + HBase/NoSQL

Dataset: 1.06M Transactions + Live Shopify Stores **Target:** Executive Review / Demo / Technical Defense

1. What the Project Does: Core Architecture & Functionality

The **Retail Customer Intelligence Platform** is an enterprise-grade Big Data and Machine Learning system that ingests high-volume retail transactions and live commercial catalogs, aggregates customer purchasing behavior, and surfaces predictive retention intelligence through an interactive executive dashboard with sub-millisecond latency.

End-to-End Pipeline in One Sentence

It turns **1,067,371 raw transactional receipts** and **live crawled Shopify product catalogs** into **customer-level behavioral cohorts, 90-day lifetime value forecasts, and proactive churn risk scores**, served in **under 1 millisecond** via a luxury executive command center.

1. Big Data Engineering (Batch Layer)

HDFS • PySpark • Hive

Cleans, deduplicates, and sanitizes 1.06M transaction records across 5,878 real customers. Eliminates cancellations (prefix `C`), negative quantities, and uncredited refunds. Aggregates multi-year histories into Snappy-compressed columnar Parquet tables with **8.4× storage compression**.

2. Machine Learning Engine

Spark MLlib • Scikit-Learn

Trains 3 validated predictive models:
- **K-Means Clustering:** Segments customers into 4 behavioral cohorts (Champions, Loyal, Potential, At-Risk).
- **CLV Regressor (Gradient Boosting):** Forecasts 90-day future spend (R^2=0.5238, MAE=$321).
- **Churn Classifier (Random Forest):** Predicts defection probability (ROC-AUC=0.8056, F1=0.7876).

3. Low-Latency Serving Layer

HBase Column Families • SQLite KV

Solves the *Dual-Speed Big Data problem*. Instead of scanning massive analytical tables on every web request, pre-computed features and inference results are materialized into structured column families (`cf_rfm`, `cf_pred`, `cf_hist`, `cf_bench`), guaranteeing **0.84ms p99 query latency**.

4. Executive Interface & Simulation

DashFlow UI • REST API v1

An emerald-themed executive dashboard featuring direct customer search, multi-store churn matrices across 5 major DTC brands, an interactive **What-If Scenario Simulator**, and automated retention playbooks with CSV data exports.

2. Why Do We Need It? The Strategic Business Case

CAC Crisis

+220%

Acquisition costs have tripled over the last 5 years. Retaining an existing customer is 5× to 7× cheaper than acquiring a new one.

Silent Churn

68%

Of lost customers defect silently without filing a complaint or survey response, leaving traditional CRM reports unaware until months later.

Discount Fatigue

$1.2M+

Wasted annually on blanket coupon blasts sent to loyal customers who would have happily purchased at full margin anyway.

> **The Fundamental Problem: Historical Reporting vs. Predictive Action**
>
> Traditional retail dashboards are **rearview mirrors**—they summarize who bought last quarter. This platform is a **predictive GPS**—it alerts growth teams which customers are in danger of lapsing *right now*, predicts how much they will spend over the next 90 days, and prescribes the exact intervention needed to retain them.

Key Business Advantages:

- **Vertical-Specific Churn Benchmarking:** E-commerce churn dynamics differ radically by industry. A 45-day gap is normal for durable footwear (*Allbirds*) but represents emergency churn for daily consumables (*Bulletproof*). Crawling live Shopify catalogs enables category-adjusted retention thresholds.
- **Pre-Campaign Scenario Simulation:** Marketers can test strategy before spending marketing budget. Adjusting recency by -15 days or frequency by +1 order instantly simulates the portfolio-wide revenue impact and CLV delta.
- **Instant Personalization at Scale:** Customer Service and VIP Account Managers query any customer record in 0.84 milliseconds, viewing their predicted CLV, lifetime order ledger, and behavioral status during live customer touchpoints.

3. How to Present It: The Executive Pitch & Live Demo Playbook

THE 30-SECOND EXECUTIVE HOOK (MEMORIZE & DELIVER FIRST)

"Customer acquisition costs have skyrocketed across retail, yet most brands let their most profitable customers quietly defect because traditional reporting is reactive. We built the **Retail Customer Intelligence Platform**—a production Big Data and AI system that processes over one million transactions and live Shopify catalogs. It predicts customer lifetime value and flags churn risk before it happens, serving rich customer intelligence to marketing leaders in under one millisecond."

The 5-Step Live Demonstration Walkthrough:

STEP & TIME	DASHBOARD SCREEN	ACTION TO PERFORM	KEY TALKING POINT
Step 1 (0:00 – 0:45)	**Executive Overview Tab** Top KPIs & Revenue Trends	Point out Gross Revenue ($2.23M), AOV ($373.13), and the monthly historical volume trajectory.	"Notice the high-level health metrics aggregated across our 1.06M transaction baseline. Everything is rendered from clean, verified columnar warehouse data."
Step 2 (0:45 – 1:45)	**Customer Command Center** 3-Column Lookup Layout	Click customer pills (`#15485` Champion, `#13748` At-Risk) or type an ID. Show the ledger and journey chart.	"Observe the response speed: **under 1 millisecond** point-lookup to our columnar NoSQL serving store. Instant profile, CLV prediction, churn risk, and complete transaction history."
Step 3 (1:45 – 2:45)	**Shopify Multi-Store Churn Intelligence**	Switch store dropdown or view the risk matrix cards (*Bulletproof, ColourPop, Allbirds, UNTUCKit, Taylor Stitch*).	"We integrate live crawled Shopify catalogs. Notice how churn risk varies by category: 36.1% for consumable coffee vs. 65.2% for high-consideration heritage menswear, with bespoke retention triggers for each."
Step 4 (2:45 – 3:45)	**Scenario Simulator & Playbook**	Drag the Recency, Frequency, and Monetary sliders. Watch the CLV number and churn badge update in real-time.	"This is our interactive What-If simulator. Marketing teams can model the exact ROI of decreasing dormancy by 20 days before funding discount campaigns."
Step 5 (3:45 – 4:30)	**Architecture Pipeline Modal & PDF**	Click the *Architecture Pipeline* button in the header to open the modal. Show the 6-stage pipeline.	"Under the hood, this is a decoupled Lambda Big Data architecture: Scraper → HDFS → Hive/Spark ETL → ML Feature Store → HBase/SQLite → Flask REST API → Edge CDN."

4. Executive Q&A Defense: Anticipated Questions & Answers

Q1: "Why did you choose a dual-layer architecture (HDFS/Hive + HBase/SQLite) instead of querying a single database?"

Answer: "Analytical queries (aggregating 1M+ rows for RFM and training ML models) require distributed columnar processing engines like Spark and Hive, which take minutes to run. Web dashboards require sub-millisecond responses. Querying analytical tables directly would freeze the UI. By pre-materializing analytical outputs into an HBase NoSQL columnar key-value store, we guarantee **0.84ms p99 query latency** without putting load on the Big Data cluster."

Q2: "How do you define and evaluate customer churn?"

Answer: "We establish a 90-day observation window. A customer is labeled as churned if they had confirmed transactions in the feature window but zero purchases during the 90-day holdout period. Our Random Forest classifier evaluates recency, frequency, monetary velocity, and tenure, achieving a balanced **ROC-AUC of 0.8056** and **F1-score of 0.7876**."

Q3: "How is this different from out-of-the-box Shopify analytics?"

Answer: "Shopify's default analytics only provide backward-looking aggregate reports. They don't provide predictive CLV forecasting, machine learning cohort clustering, what-if campaign simulation, or cross-brand market price benchmarking. Our platform gives executives predictive foresight and actionable prescriptive playbooks."

5. Production Metrics & Technical Fact Sheet

METRIC CATEGORY	SPECIFICATION / VERIFIED BENCHMARK	TECHNICAL RATIONALE
Raw Data Scale	`1,067,371` rows • `5,878` unique customers	UCI Online Retail II two-year transactional dataset.
Shopify Crawled SKUs	`2,038` live commercial products crawled	Allbirds, ColourPop, Bulletproof, UNTUCKit, Taylor Stitch.
Indexed Profiles	`5,942` pre-indexed Key-Value customer rows	HBase / SQLite embedded NoSQL serving store.
Serving Query Latency	**0.84 ms** (p99 latency benchmark)	Direct primary key lookup across indexed column families.
CLV Prediction Engine	Gradient Boosting Regressor • **R^2 = 0.5238** • MAE = $321.78	Winsorized at 99th percentile ($6,245.56 cap) to prevent outlier distortion.
Churn Classifier	Random Forest • **ROC-AUC = 0.8056** • F1 = 0.7876	Balanced class weights across 90-day holdout observation window.
Storage Optimization	**8.4× compression** via Snappy Parquet	Columnar dictionary encoding and partitioned analytical directory layout.
Automated Test Coverage	**35/35 Passing Core Tests** (API, Integrity, Platform)	Zero regression testing across serving, ML bounds, and UI synchronization.
Production Deployment	Dual-Execution: **Render** (WSGI) + **Vercel** (Edge CDN)	Decoupled frontend CDN routing API requests to Gunicorn WSGI cluster.